Software Design Document (M3)

Community Waste Reporting and Collection Tracking System ("ชุมชนสะอาด")

Team Larb Lamphoon · Introduction to Software Engineering (15031001) · Design milestone · v1.0

Repo: https://github.com/Adichat-dev/Larb-Lamphoon.git

> **Draft status:** Designed to follow SRS v1.0 (M2) exactly: browser-only app, no accounts, simulated notifications. Field names, function names and screens are proposed from the SRS and must be checked against the prototype before submission.

1. Introduction

1.1 Purpose

This document describes how the system specified in the SRS (M2, v1.0) is designed and built: its architecture, data model, interfaces, main flows, screens and security measures. Every design element traces back to a functional requirement (FR-1 to FR-5) or a non-functional requirement (NFR-1 to NFR-5).

1.2 Scope

Citizens report waste spots and track them by tracking number. Collection officers work a queue sorted by urgency. Administrators view summaries and export CSV. Notifications are simulated as on-screen messages.

2. Constraints Inherited from the SRS

Every design decision below follows the assumptions, constraints and out-of-scope list of SRS v1.0. The design adds nothing the SRS excludes.

SRS v1.0 says	Design consequence
2.2: prototype uses simulated data only	Demo reports are seeded in the browser; no real personal data is collected
2.2: all data is stored in the user's browser (local storage), not shared between devices	No server or database; every module reads and writes browser local storage
2.2: notifications are on-screen messages, not sent to phones	The notification module shows a message on screen; nothing is sent
2.2: user interface is in Thai	All labels, buttons and messages are in Thai
2.1 and 6: no accounts, login or role-based access control	No registration or login; the user picks a role view (citizen, officer, administrator) from a switcher
2.1 and 6: no truck routing, treatment, recycling or fines	None of these is designed

3. System Architecture

3.1 Style

A client-only, single-page web application: all logic runs in the browser and all data stays in the browser's local storage, as SRS 2.2 requires.

Layer	Technology	Responsibility
Presentation	HTML, CSS and JavaScript, Thai UI	Renders the citizen, officer and administrator views; responsive at 360 px
Application logic	JavaScript modules running in the browser	Validation, tracking-number generation, queue ordering, status changes, summaries, CSV generation

Layer	Technology	Responsibility
Data	Browser local storage	Keeps reports, status history, photos and notifications on the user's device

3.2 Component overview

Component	Role	Realises
Role switcher	Switches between citizen, officer and administrator views	NFR-1 (switch role view), UC-1 to UC-5
Report module	Validates the form, creates the report with status Pending, assigns the tracking number	FR-1, FR-2
Photo module	Accepts an image file and saves it with the report	FR-1, FR-3
Tracking module	Finds one report by its exact tracking number and shows its details and history	FR-2, NFR-4
Job queue module	Lists Pending and In progress reports by urgency, then age; accepts and completes jobs	FR-3
Notification module	Shows an on-screen message when a status changes	FR-4
Dashboard and export module	Counts by status, repeated-location list, CSV download	FR-5
Storage module	Reads and writes local storage so data survives a refresh	NFR-3

3.3 Deployment

The app is a set of static files (HTML, CSS, JavaScript) opened in a smartphone or computer browser with an internet connection, as SRS 2.2 assumes. No server or database is needed. Because data lives in each browser, a report made on one device is not visible on another.

4. Data Design

4.1 Entities

Report

Field	Type	Notes
trackingNo	text, unique	Format WS-0001, from a running counter
wasteType	text	Chosen from a list
urgency	text	Chosen from a list (for example low, medium, high)
description	text, required	FR-1 criterion 2
location	text, required	Area or address; the GPS button may fill in coordinates
status	text	Pending, In progress or Collected
createdAt	datetime	Report time, also used in the CSV
photos	list of Photo	Optional citizen photo and optional officer after-photo
history	list of StatusEntry	Dated status history

StatusEntry (stored inside each Report)

Field	Type	Notes
status	text	Pending, In progress or Collected
time	datetime	Time stamp shown in the dated history (FR-2, FR-3)

Photo (stored inside each Report)

Field	Type	Notes
data	text	Image saved as a data URL in local storage
kind	text	before (citizen) or after (officer)

Notification

Field	Type	Notes
trackingNo	text	The report concerned
status	text	The new status
message	text	Names the tracking number and new status (FR-4 criterion 1)
time	datetime	When the change was saved

4.2 Relationships

- One Report holds many StatusEntry and Photo records inside itself.
- Each Notification refers to one Report by trackingNo.
- A counter in local storage gives the next tracking number.
- The repeated-area list groups reports by the location value, so the form should offer area names consistently.
- Local storage holds only a few megabytes, so photos are shrunk before saving.

4.3 Status lifecycle

Pending → In progress (officer accepts) → Collected (officer marks collected). Collected jobs leave the open list. No backward transitions are allowed.

5. Interface Design

With no server, the interfaces are the screen actions and the functions behind them. Function names are proposed and should match the prototype's code.

Action	Function (proposed)	Used by	FR
Submit report	createReport(form)	Citizen	FR-1
Search tracking number	findReport(trackingNo)	Citizen	FR-2
List open jobs	listOpenJobs()	Officer	FR-3
Accept job	acceptJob(trackingNo)	Officer	FR-3
Mark collected	completeJob(trackingNo, afterPhoto)	Officer	FR-3
Show notification	notifyStatusChange(report)	System	FR-4
Dashboard figures	getSummary()	Administrator	FR-5
Export CSV	exportCsv()	Administrator	FR-5

Validation errors appear as Thai messages beside the form field. An unknown tracking number shows a "not found" message. Every action updates the screen within 10 seconds (NFR-1).

6. Behaviour Design (Main Flows)

UC-1 Report waste spot (FR-1)

1. The citizen opens the report form and fills in waste type, urgency, description and location (typed or from the GPS button), with an optional photo.
2. The citizen submits the form.

3. The report module checks that description and location are filled in. If either is empty, the form shows which is missing and nothing is saved.
4. Otherwise the app saves the report with status Pending and a first history entry, and saves the photo with it.
5. The screen shows the new tracking number (for example WS-0001).

UC-2 Track report (FR-2)

1. The citizen enters a tracking number.
2. The tracking module looks for exactly that number.
3. If found, the screen shows status, urgency, type, location, description, photos and dated history, and nothing about other reports or people. If not, it shows "not found".

UC-3 Update collection status (FR-3)

1. The officer opens the job list: Pending and In progress reports, most urgent first and oldest first within the same urgency.
2. The officer accepts a Pending job: its status becomes In progress and a time-stamped history entry is added.
3. The officer marks an In progress job collected, optionally attaching an after-photo: its status becomes Collected, a history entry is added, and the job leaves the list.

UC-4 Receive status notification (FR-4)

1. When the officer changes a status, the same action creates a notification.
2. The screen displays a message naming the tracking number and the new status. It is simulated: nothing is sent to a phone.
3. When the citizen later looks up the tracking number, the new status and time already appear in the history.

UC-5 View dashboard and export CSV (FR-5)

1. The administrator opens the dashboard and sees the total number of reports and the number in each status.
2. Reports are grouped by location; locations reported more than once are listed with their counts.
3. The administrator exports: a CSV file downloads with tracking number, area, waste type, urgency, description, status and report time for every report.

7. User Interface Design

The interface is in Thai and runs as one page; a role switcher changes the view between citizen, officer and administrator.

Screen	Role	Main elements
Report form	Citizen	Waste type, urgency, description, location field, GPS button, photo picker, submit
Tracking	Citizen	Tracking number box, result card with dated status history, "not found" message
Job list	Officer	Cards ordered by urgency, Accept and Mark collected buttons, after-photo picker
Dashboard	Administrator	Totals by status, repeated-area list, Export CSV button

Screen	Role	Main elements
Notification	Citizen, officer	On-screen message with tracking number and new status

Layouts are responsive and need no horizontal scrolling at 360 px width (NFR-5). The report form is one short screen so a first-time citizen can finish it in under 90 seconds (NFR-2).

8. Security Design

Concern	Design decision	Requirement
Lookup privacy	A report is shown only for its exact tracking number; the citizen view has no list of reports and no search by name or area	NFR-4
Personal data	The report form has no name, phone or e-mail field, so no personal data is stored	NFR-4
Data location	Data stays in the user's own browser and is never sent to a server	SRS 2.2
Input handling	Required fields are checked, only image files are accepted as photos, and user text is shown as plain text, not HTML	FR-1
CSV safety	Cells starting with =, +, - or @ are neutralised on export	FR-5
Role views	The role switcher is a convenience, not access control; the SRS excludes login and role-based access control	SRS 6

9. Traceability Matrix

Requirement	Design element	Test
FR-1	Report module, photo module, report form	Submit-report test (valid, missing description or location, with photo)
FR-2	Tracking module, history timeline	Tracking lookup test (valid and unknown number)
FR-3	Job queue module, status transitions	Queue order and status test
FR-4	Notification module	Notification test
FR-5	Dashboard and export module	Dashboard and CSV test
NFR-1	All actions run locally in the browser	Built-in response-time panel, 3 trials per action
NFR-2	One short report form	Timed user test with 3 people
NFR-3	Storage module writes to local storage on every change	Refresh and reopen test
NFR-4	Section 8	Search with a wrong number; inspect the citizen view
NFR-5	Responsive layout	Resize browser to 360 px; test on one phone

10. Known Limitations

- Data is stored per browser, so reports are not shared between devices; a citizen, an officer and an administrator on different devices do not see the same data.
- Notifications are on-screen only; nothing reaches a citizen's phone.
- There are no accounts, login or access control.
- The prototype uses simulated data; real use would need a server and database.
- Local storage has a small size limit, which limits how many photos can be kept.

11. AI Usage Log

Date	Tool	What we asked	What AI produced	What we changed or verified	What we learned
8 Nov 2026	Claude	Write the M3 design document from SRS v1.0, then revise it to follow the SRS exactly	Drafted all sections, then rewrote constraints, data, interfaces, flows and security to match the SRS	**Team to fill in:** check field names, function names and screens against the prototype; confirm M3's required sections	**Team to fill in**